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THE HUMAN FIREWALL

DIGITAL SAFETY STARTS WITH YOU

INTRODUCTION TO CYBERSECURITY

Key stat

95% of cybersecurity breaches are caused by human error.

What is cybersecurity?

Cybersecurity is the practice of protecting systems, networks, and data from digital attacks, theft, and damage. It involves technologies, processes, and awareness training that help organizations and individuals defend against threats like phishing, malware, and hacking. In the context of phishing awareness, cybersecurity emphasizes recognizing suspicious emails or messages, avoiding malicious links or attachments, and safeguarding sensitive information to prevent attackers from gaining unauthorized access.

Why it's more relevant than ever

- Rising phishing attacks: Scammers use convincing emails and messages to trick people into revealing sensitive data.
- High financial impact: Breaches can cost companies millions in recovery, fines, and lost trust.
- Global scale: Cyberattacks can disrupt governments, supply chains, and critical infrastructure.

THE HUMAN ELEMENT

IN CYBERSECURITY

People as the first line of defense

Think of cybersecurity like a castle. The firewalls, encryption, and security software are the stone walls and towers, but the guards at the gate—the people—are the first to spot intruders trying to sneak in. If the guards let someone disguised as a friend walk through without checking, the whole castle is at risk. That's why employees, through vigilance and awareness, act as the true first line of defense

Common attack vectors that target humans

Phishing, social engineering, weak passwords

Cybersecurity isn't just IT's responsibility, It's all our responsibility

UNDERSTANDING

SOCIAL ENGINEERING

Definition and examples

Definition:Social engineering is when attackers manipulate people into revealing information or taking actions that compromise security.

Example:A phishing email that looks like it's from your bank, asking you to "verify" your account by clicking a link and entering your login details.

How attackers manipulate trust and emotions

They often pretend to be trusted figures—like banks, employers, or colleagues—to gain credibility. At the same time, they use emotional triggers to push people into acting quickly without thinking:

Fear: Warning that your account will be locked or you'll lose money unless you act immediately.

Empathy/Helpfulness: Appealing to kindness, such as pretending to be a coworker in need of assistance.

Real-world case study

The 2011 RSA breach was a landmark cyberattack in which hackers used a spear phishing email with a malicious Excel attachment exploiting a zero-day Adobe Flash vulnerability to infiltrate RSA's network and deploy the Poison Ivy malware. Once inside, they escalated privileges and accessed sensitive data tied to RSA's SecurID two-factor authentication tokens, widely used by government agencies and defense contractors. The incident compromised the security of over 40 million users, cost RSA's parent company EMC more than $66 million in remediation, and exposed the vulnerabilities of even leading cybersecurity firms. Ultimately, it highlighted the critical need for employee awareness, stronger defenses against advanced persistent threats, and vigilance

Security Is A Human Layer: RSA has advanced security tools, but the human element-the decision to open the an attachment was the critical vulnerability. This highlights that technology alone isn't enough in protecting sensitive data.

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What phishing looks like

Phishing is a cyber-attack that disguises itself as a legitimate message—often an email, text, or website—to trick you into revealing sensitive information. These messages may appear to come from trusted organizations like banks, delivery services, or government agencies. They often use official-looking logos, urgent language such as "Your account will be locked", and links that lead to fake websites designed to steal your data.

Types

Email, SMS (smishing), voice (vishing)

THE MOST COMMON THREAT

PHISHING

How to spot red flags

Key warning signs include:

- Unexpected requests for personal or financial information
- Suspicious or misspelled links and email addresses
- A sense of urgency or pressure to act quickly

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PASSWORDS &
AUTHENTICATION

BEST PRACTICES

Dangers of weak or reused passwords

Using weak or repeated passwords makes it easy for attackers to break into accounts. If one account is compromised, reused passwords allow criminals to access many others, leading to stolen data, identity theft, or financial loss. Strong, unique passwords are essential to protect both personal and organizational information.

Importance of password managers

Password managers are powerful tools that help you create, store, and manage strong, unique passwords for every account. Instead of relying on memory or writing them down, a password manager securely encrypts your credentials and automatically fills them in when needed.

Enabling multi-factor authentication (MFA)

Multi-Factor Authentication (MFA) is a security measure that requires more than just a password to access an account, adding an extra layer of protection. It works by combining something you know (your password) with something you have (like a code sent to your phone or an authenticator app) or something you are (like a fingerprint). To implement MFA, go to the security settings of your account, enable options such as “Two-Step Verification” or “Multi-Factor Authentication,” and follow the prompts to link your phone, app, or device. This way, even if your password is stolen, attackers cannot access your account without the second factor.

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SAFE ONLINE

BEHAVIOR

Think before you click

Phishing often relies on impulsive clicks. Always pause before opening links or attachments in emails, texts, or pop-ups. If something feels urgent or suspicious, verify it first—clicking without thinking can expose your device and data to attackers.

Verify sources and links

Phishing messages often mimic trusted organizations. Double-check the sender's email address, look for spelling errors, and hover over links to see the actual destination before clicking. If in doubt, go directly to the official website rather than relying on embedded links.

Use secure Wi-Fi networks

Public Wi-Fi may be convenient, but it's often unsecured and vulnerable to hackers. Avoid logging into sensitive accounts or entering personal information on open networks. Instead, use trusted, password-protected Wi-Fi or a VPN to keep your connection safe.

Be cautious with downloads and attachments

Be Cautious with Downloads and Attachments. Attachments and downloads are common ways attackers deliver malware. Only open files from trusted sources, and be wary of unexpected attachments—even if they appear to come from colleagues or familiar companies. When unsure, confirm with the sender before opening.

REPORTING AND RESPONSE

Why early reporting is critical

- Stops attacks before they spread
- Limits damage and protects sensitive data
- Gives security teams time to respond effectively
- Prevents attackers from gaining deeper access

How to report suspicious activity in your organization

- Follow your company's reporting process (e.g., "Report Phishing" button or IT contact)
- Forward suspicious emails to the security team
- Never ignore or delete suspicious messages without reporting
- Confirm with IT if you're unsure how to proceed

Role of incident response teams

- Investigate and contain security threats
- Analyze phishing attempts and remove malicious content
- Strengthen defenses to prevent future attacks
- Act quickly to protect individuals and the organization

BUILDING A CULTURE

OF CYBER AWARENESS

Regular training and simulations

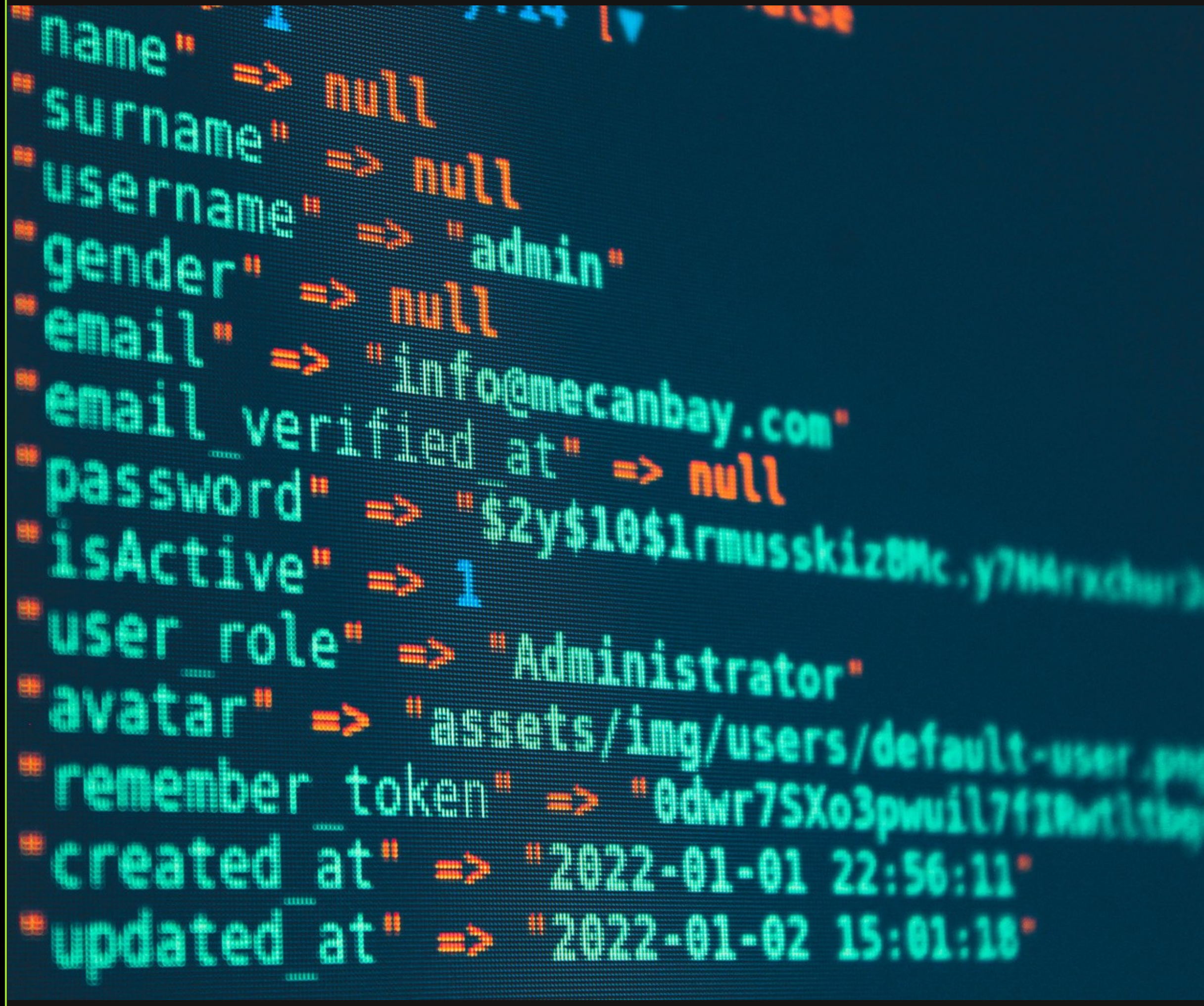
- Ongoing training keeps employees alert to evolving threats
- Simulated phishing exercises build practical awareness
- Reinforces best practices through real-world scenarios
- Helps identify areas where extra support is needed

Encouraging peer accountability

- Every individual plays a role in cybersecurity
- Report suspicious activity promptly and responsibly
- Follow organizational policies and security guidelines
- Understand that personal vigilance protects the whole team

Cybersecurity as part of daily workflow

- Treat security as a routine part of work, not an extra task
- Use strong passwords, MFA, and secure networks consistently
- Verify links, attachments, and sources before engaging
- Make safe online behavior a daily habit across all tasks



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FINAL THOUGHTS

& CALL TO ACTION

Recap

You are the human firewall

One small mistake can lead to a big breach

Take action today

Update passwords, stay informed, be alert

Remember, Digital safety starts with you

SANS Institute – Phishing Awareness Training
Offers professional training modules, simulations, and articles to help organizations build resilience against phishing.
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